

Week-8:

Experiment 3: Password Strength Checker

Aim: To implement form handling and validation logic.

Description:

This component checks password strength based on rules and displays feedback to the user.

Procedure:

1. Create a component PasswordChecker.
2. Add an input field for password.
3. Capture input using useState.
4. Apply validation rules:
 - Minimum length (8)
 - Uppercase letter
 - Number
 - Special character
5. Based on conditions, display:
 - Weak
 - Medium
 - Strong
6. Show result dynamically as user types.

Outcome:

- Learn **form handling**
- Understand **real-time validation**
- Apply logical conditions

Code:

Password.jsx:

```
import React, {useState} from "react";  
  
function Password(){  
  const [password,setPassword]=useState("");  
  
  const [strength,setStrength]=useState("");
```

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```
const checkStrength=(pass)=> { let score=0;

  if(pass.length>=0)      score++;

  if(/[A-Z]/.test(pass))    score++;

  if(/[0-9]/.test(pass))    score++;

  if(/[!@#$$%^&*]/.test(pass))    score++;

  if(score<=2)      setStrength("Weak");

  else if(score==3)  setStrength("Medium");

  else      setStrength("Strong"); };

const handleChange=(e)=> { const value=e.target.value;

  setPassword(value);  checkStrength(value); }

return( <div style={{textAlign:"center",marginTop:"50px"}}>

  <h1>Password Checker</h1>      <input      type="password"

  placeholder="Enter Password"    value={password}

  onChange={handleChange}/>      <h2>Strength: {strength}</h2> </div>

);}export default Password;
```

App.jsx:

```
import React from "react";import Password from "./Password";

function App() {

  return ( <div>    <Password />  </div> );

}export default App;
```

Output:

Password Checker

.....

Strength: Weak

Experiment 4 : Multi-Page App using React Router

Aim: To understand **routing and navigation in React**.

Description: This experiment creates a multi-page React app with navigation using React Router.

Procedure:

1. Install router:

```
npm install react-router-dom
```

2. Create components:

- Home
- About
- Contact

3. Configure routes using BrowserRouter.

4. Add navigation bar with links.

5. Use Route and Link components.

6. Navigate between pages without reload.

Outcome:

- Understand **Single Page Applications (SPA)**
- Learn **routing concepts**
- Build multi-page structure

Code:

Home.jsx:

```
function Home()  
{ return <h2>Home Page</h2>  
export default Home;}
```

About.jsx:

```
function About()  
{ return<h2>About Page</h2>;}  
export default About;
```

Contact.jsx:

```
function Contact()  
{ return <h2>Contact page</h2>;  
}export default Contact;
```

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App.jsx:

```
import React from "react";
import Password from "../Password";
import {BrowserRouter,Routes,Route,Link} from "react-router-dom";
import Home from "../Home";
import About from "../About";
import Contact from "../Contact";
function App()
{ return( <BrowserRouter>
<div style={{textAlign:"center",}}>
<h1>React Router</h1>
<nav>
<Link to="/">Home</Link>|{" "}
<Link to="/about">About</Link>|{" "}
<Link to="/contact">Contact</Link>
</nav>
<Routes>
<Route path="/" element={<Home/>}/>
<Route path="/about" element={<About/>}/>
<Route path="/contact" element={<Contact/>}/>
</Routes>
</div>
</BrowserRouter>
);
}
export default App;
```

Output:



Experiment 5: To understand and implement **Props in React** by passing data from a parent component to child components and rendering dynamic UI.

Create a React application that displays **student profiles** using props.

Each student card should display:

- Name
- Branch
- Year
- Profile Image

Props (Properties) are used to pass data from a parent component to a child component.

Props are read-only (immutable).

They help in making components reusable and dynamic.

Example:

```
<Greeting name="Shiva" />
```

Code:

1. StudentCard.jsx (Child Component):

```
function StudentCard(props) {  
  return (

border: "1px solid black",  
    padding: "15px",  
    margin: "10px",  
    width: "200px",  
    borderRadius: "10px"  }}>  
    <img src={props.image} alt="profile" width="100%" />  
    <h3>{props.name}</h3>  
    <p>Branch: {props.branch}</p>


```

<p>Year: {props.year}</p>

</div>);}

export default StudentCard;

2. App.jsx (Parent Component):

import React from "react";

import StudentCard from "./StudentCard";

function App() { return (

<div style={{ textAlign: "center" }}>

<h1>Student Profiles</h1>

<div style={{ display: "flex", justifyContent: "center" }}>

<StudentCard name="Vijitha" branch="CSE" year="2nd Year"

image="https://via.placeholder.com/150" />

<StudentCard name="Soumya" branch="ECE" year="3rd Year"

image="https://via.placeholder.com/150" />

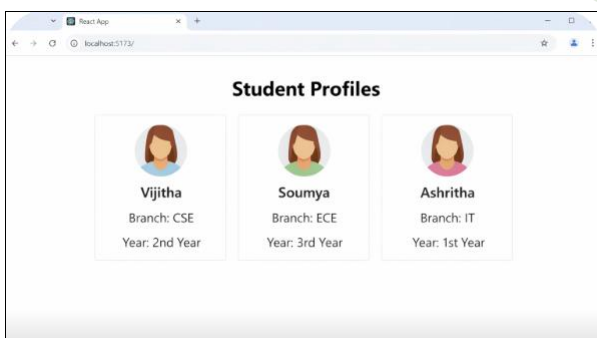
<StudentCard name="Ashritha" branch="IT" year="1st Year"

image="https://via.placeholder.com/150" /> </div> </div>);

}

export default App;

Output:



Experiment 6: Student Table with Search & Pagination

Aim: To implement **data rendering, filtering, and pagination**.

Description: Displays a list of students in a table with search and pagination features.

Procedure:

1. Create a component StudentTable.
2. Create sample student data (array of objects).
3. Display data using a table.
4. Add search input:
 - Filter students by name
5. Implement pagination:
 - Show 5 records per page
 - Add Next/Previous buttons
6. Use useState for:
 - current page
 - search query
7. Update UI dynamically.

Outcome:

- Understand **props & data flow**
- Learn **list rendering**
- Implement **pagination logic**

Code:

1.StudentTable.jsx:

```
import React, { useState } from "react";  
function StudentTable() {  
  const students = [  
    { id: 1, name: "Vijitha", branch: "CSE", year: "2nd" },  
    { id: 2, name: "Soumya", branch: "ECE", year: "3rd" },  
    { id: 3, name: "Ashritha", branch: "IT", year: "1st" },  
    { id: 4, name: "Ravi", branch: "CSE", year: "4th" },  
    { id: 5, name: "Kiran", branch: "EEE", year: "2nd" },  
    { id: 6, name: "Anil", branch: "MECH", year: "3rd" },  
    { id: 7, name: "Pooja", branch: "CIVIL", year: "1st" },  
    { id: 8, name: "Sneha", branch: "CSE", year: "2nd" },  
  ]  
}
```


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```
{ id: 9, name: "Rahul", branch: "ECE", year: "4th" },
{ id: 10, name: "Divya", branch: "IT", year: "3rd" } ];
const [search, setSearch] = useState("");
const [currentPage, setCurrentPage] = useState(1);
const recordsPerPage = 5;
const filteredStudents = students.filter((student) =>
  student.name.toLowerCase().includes(search.toLowerCase()) );
const indexOfLast = currentPage * recordsPerPage;
const indexOfFirst = indexOfLast - recordsPerPage;
const currentStudents = filteredStudents.slice(indexOfFirst, indexOfLast);
const totalPages = Math.ceil(filteredStudents.length / recordsPerPage);
return (
<div style={{ textAlign: "center" }}>
  <h1>Student Table</h1>
  <input type="text" placeholder="Search by name"
    value={search} onChange={(e) => {
      setSearch(e.target.value); setCurrentPage(1);
    }} />
  <table border="1" style={{ margin: "20px auto" }}>
    <thead>
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Year</th>
      </tr>
    </thead>
    <tbody>
      {currentStudents.map((student) => (
        <tr key={student.id}>
          <td>{student.id}</td>
          <td>{student.name}</td>
          <td>{student.branch}</td>
          <td>{student.year}</td>
        </tr>
      ))}
    </tbody>
  </table>
  <button
    onClick={() => setCurrentPage(currentPage - 1)}
    disabled={currentPage === 1} > Previous
```



```
</button>
  <span style={{ margin: "0 10px" }}>
    Page {currentPage} of {totalPages}
  </span>
</button>
  onClick={() => setCurrentPage(currentPage + 1)}
  disabled={currentPage === totalPages} > Next </button>
</div> );}
export default StudentTable;
```

3. App.jsx:

```
import React from "react";
import StudentTable from "./StudentTable";
function App()
{
  return (
    <div>
      <StudentTable />
    </div> );}
export default App;
```

Output:

